

RollNo.

--	--	--	--	--	--	--	--

ODD END SEMESTER EXAMINATION DECEMBER – 2025

Name of the Course: B.Tech

Semester: 5

Name of the Paper: Electrical Machine Design

Paper Code: TEE-504

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20marks)	
(a)	Explain the major considerations in the design of electrical machines.	CO1
(b)	Write short notes on magnetic leakage and its effect on performance of machines.	
(c)	Explain the difference between induced and forced ventilation techniques with suitable diagrams.	
Q2	(20marks)	
(a)	Explain the cooling methods in transformer.	CO2
(b)	Discuss the impact of temperature rise in transformer.	
(c)	Explain the design of transformer core.	
Q3	(20marks)	
(a)	Explain the construction of field system and rotor system of DC machine.	CO3
(b)	Discuss the selection of poles for the DC machine.	
(c)	How does the length of air gap impacts DC machine. Explain	
Q4	(20marks)	
(a)	Explain the equivalent resistance in squirrel cage induction motor.	CO4
(b)	Write short notes on a) Design of stator windings b) Design of rotor windings.	
(c)	Explain the temperature rise and efficiency in three phase induction motor.	
Q5	(20marks)	
(a)	Explain the classification of alternators.	CO5
(b)	Discuss the specific loading for the alternators.	
(c)	Explain the design of stator winding.	

